

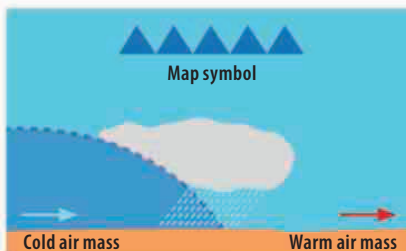
# Weather

CHANGES IN CONDITIONS IN THE ATMOSPHERE PRODUCE DIFFERENT WEATHER EVENTS.

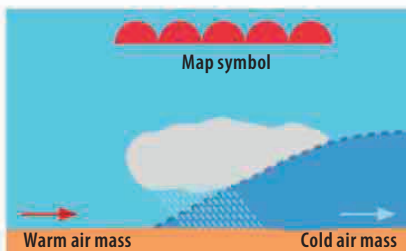
Weather changes occur when sections of atmosphere with different temperatures, pressures, and humidities (water content) come into contact.

## Precipitation

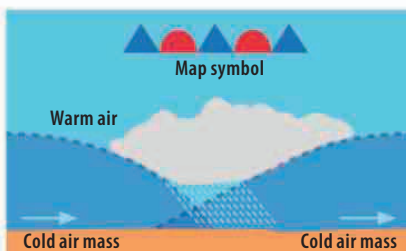
Rain is an example of precipitation, where water vapor in the atmosphere condenses to a liquid and falls to the ground. Warm air can hold more water vapor than cool air. Precipitation occurs when air saturated with water vapor is forced to cool and the excess falls as raindrops. Hail and snow are also forms of precipitation. Hailstones are formed when rain is repeatedly blown upward into colder areas of the atmosphere, while snowflakes form when water vapor condenses in already freezing air. Precipitation occurs at weather fronts, where air masses of different temperatures meet. There are three types, shown below.



◁ **Cold front**  
Cold air moves under warmer, wetter air. As the warm air rises, its pressure falls and it cools, dropping its water as rain.



◁ **Warm front**  
A mass of warm air flows over a block of cold air, forming rain and clouds. Warm fronts move more slowly than cold fronts, resulting in sustained rain.



◁ **Occluded front**  
This occurs when the warm air mass is pushed off the ground completely by cooler air. Occluded fronts also produce rain.

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## Wind

Wind forms when air rushes from an area of high pressure to an area of low pressure. The bigger the difference between the two pressures, the stronger the wind.

### ▽ Beaufort wind scale

This scale describes the strength of wind by its effects, so people can judge wind speeds without a measuring device.

| Scale | Wind speed km/h (mph) | Strength        | Observation            |
|-------|-----------------------|-----------------|------------------------|
| 0     | 0–2 (0–1)             | calm            | smoke rises vertically |
| 1     | 3–6 (2–3)             | light air       | smoke drifts slowly    |
| 2     | 7–11 (4–7)            | light breeze    | leaves rustle          |
| 3     | 12–19 (8–12)          | gentle breeze   | small flags fly        |
| 4     | 20–29 (13–18)         | moderate breeze | trees toss, dust flies |
| 5     | 30–39 (19–24)         | fresh breeze    | small branches sway    |
| 6     | 40–50 (25–31)         | strong breeze   | large branches sway    |
| 7     | 51–61 (32–38)         | near gale       | trees in motion        |
| 8     | 62–74 (39–46)         | gale            | twigs break            |
| 9     | 75–87 (47–54)         | strong gale     | branches break         |
| 10    | 88–101 (55–63)        | storm           | trees snap             |
| 11    | 102–119 (64–74)       | violent storm   | widespread damage      |
| 12    | 120+ (75+)            | hurricane       | extreme damage         |

### REAL WORLD

## Tornado

The fastest winds on Earth are inside tornadoes. They form when a column of spinning air inside a thunderstorm cloud makes contact with the ground. An average tornado is about 80 m (260 ft) across and the air in it moves at 170 km/h (110 mph), sucking objects off the ground, high into the air.

